A regime aware portfolio replica strategy

Replicating a Black-Box Target Portfolio with Liquid Futures

Business Case 3 – Fintech Project

Group 6

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Outline

1. Problem & Data
2. Regime Detection Framework
3. What Didn't Work
4. Soft Markov Switching Elastic Net
5. Kalman Filter
6. Soft Bayesian Stacking
7. Risk Analysis

The Problem: Black-Box Replication with Futures

Setup

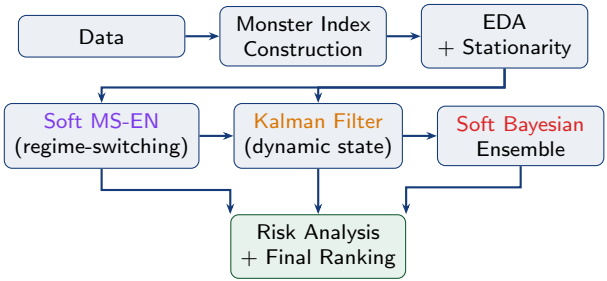
- **Target**: a synthetic diversified index (*Monster Index*) — weights unknown
- **Tools**: 10 liquid futures contracts (equity, rates, commodities)
- **Data**: Bloomberg weekly prices, Oct 2007 – Apr 2021 (705 weeks)
- **Constraint**: UCITS-style — max leverage $2\times$ or max 1M VaR(99%) $\leq 20\%$

Research Question

Can we replicate the Monster Index **purely from futures returns**, without ever observing its holdings?

Challenge

Correlations and factor exposures are **time-varying** - static models break across regimes. **Our answer**: a fully dynamic, regime-aware framework.



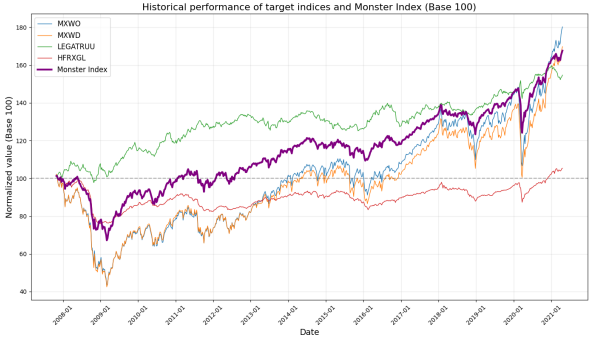
Futures Universe (10 contracts)

ES1/NQ1	US Equity	RX1/TY1/DU1/TU2	Rates
VG1/TP1	Intl Equity	GC1/CO1	Comm

LLL1 (MSCI EM) removed — stale prices after 2011.

Monster Index & Exploratory Analysis

Monster Index: Sharpe-weighted blend of 4 benchmarks

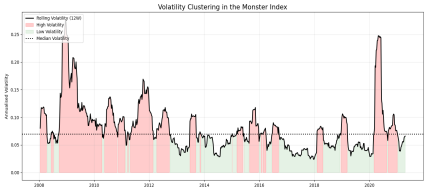


Ann.Ret	Ann.Vol	Sharpe	Max DD	Skewness	Kurtosis
4.25%	9.27%	0.46	33.71%	−0.72	5.80

EDA: Three Key Findings

- **Fat tails** — QQ-plots of all futures show systematic tail deviations from Normal
- **Volatility clustering** — Ljung-Box on *squared* returns strongly rejects H_0 at all lags ⇒ time-varying variance
- **Stationarity confirmed** — ADF & Phillips-Perron both reject unit root in returns (p-value ≈ 0)

Volatility clustering

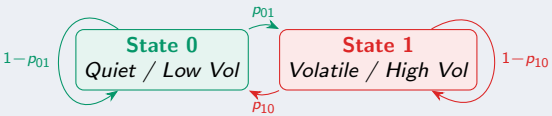


Regime Detection: Baum-Welch HMM + Online Bayesian Filter

Why Regimes Matter

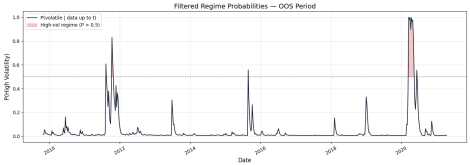
Market relationships between the Monster Index and futures **shift structurally** during crises. A static model trained on calm data will be blind to stress periods — and vice versa.

Two identified states:



Regimes are persistent: e.g. once in a crisis, the model expects it to last several months before reverting

Filtered Regime Probabilities over Time



Two-Step Estimation (No Lookahead)

1. **Baum-Welch EM** on first 104 weeks: estimates A_0, μ_0, σ_0 from data — transition probabilities are *learned*, not assumed
2. **Online Bayesian filter** (Dirichlet + Normal-Gamma): updates posteriors one week at a time $\Rightarrow$ causal $P(\text{volatile} \mid t)$

Explored Approaches That Were Abandoned

OLS / Elastic Net — Configuration Search

Idea We tested four configurations sequentially, adding complexity only when it produced a clear improvement. Starting from a plain weekly long/short OLS (R1), we moved to monthly rebalancing (R4), then added exponential sample weighting (EWM R4), and finally imposed a long-only constraint (R4 LO).

Problem No single configuration managed to satisfy all thresholds at once, and market regimes were never taken into account.

Config	IR	TEV%	Corr	Sharpe	Ret%	Vol%	MaxDD%	Avg GE	Avg TO	MSE
OLS R1	−0.081	3.80	0.862	0.88	5.68	6.47	12.7	1.47	10.06	$2.77 \cdot 10^{-5}$
OLS R4	+0.036	3.64	0.874	0.94	6.12	6.54	13.0	1.48	4.51	$2.55 \cdot 10^{-5}$
EWM R4	+0.126	3.79	0.863	0.99	6.47	6.52	11.9	1.78	5.36	$2.75 \cdot 10^{-5}$
OLS R4 LO	−0.034	3.53	0.883	0.89	5.87	6.56	12.3	0.98	2.61	$2.39 \cdot 10^{-5}$
Pooling	−0.153	3.76	0.883	0.73	4.91	6.74	11.6	0.48	0.69	$2.72 \cdot 10^{-5}$
MS-EN	−0.035	3.94	0.872	0.78	5.36	6.85	10.8	1.55	5	$2.96 \cdot 10^{-5}$

Pooling *(O'Doherty et al., 2017)*

Idea Futures were grouped into four factor pools (US equity, international equity, rates, and commodities) and combined using log-score optimal weights.

Problem Tracking error and information ratio remained worse than the plain EN on every metric.

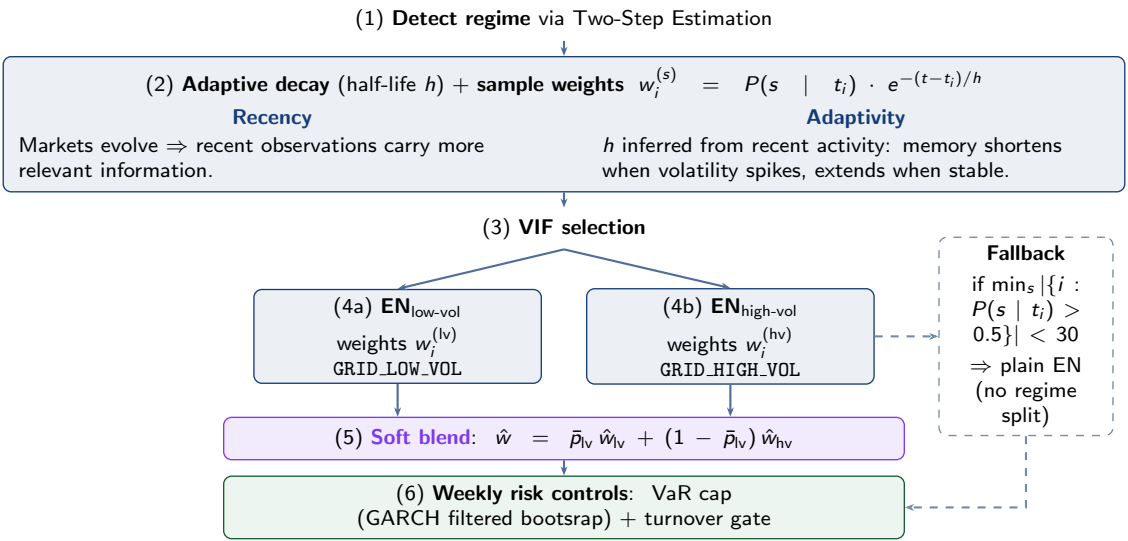
Kalman with Fixed Process Noise

Idea A standard Kalman filter where the process noise matrix Q is held constant throughout the sample.

Problem Market regimes were never taken into account

Model 1: Soft Markov Switching Elastic Net (MS-EN)

How the model works at each rebalancing step (104w; 4w reb)



Model 1: MS-EN - Results

Cumulative Performance vs Target



Why soft blending?

To emphasise low- and high-volatility regimes respectively to let factor exposures adapt dynamically as market conditions shift. Hard switching is avoided because forcing a discrete assignment discards filter uncertainty and generates discontinuous portfolio rebalancing that inflates turnover and transaction costs.

MS-EN Performance Metrics

Metric	Replica	Target
Ann. Return	**5.36%**	5.50%
Ann. Volatility	**6.85%**	8.04%
Sharpe Ratio	**0.782**	0.684
Max Drawdown	**10.8%**	15.9%

Correlation	0.8718
Information Ratio	−0.0350
Tracking Error Vol	3.94%
RMSE (weekly)	0.5460%
Ann. Turnover (1-way)	5×
Avg Gross Exposure	1.551

Model 2: Kalman Filter — State-Space Formulation

State-Space Formulation

Rather than fitting a fixed regression periodically, the Kalman Filter treats the **portfolio weights as hidden quantities** that drift over time and must be continuously inferred from observed returns.

Two layers describe the world:

- **Transition layer** — weights are assumed to change slowly and randomly each week. How much they are allowed to move is governed by the *process noise*, which the HMM sets each week based on the current regime.
- **Observation layer** — the target return each week is assumed to equal the dot product of current futures returns and the (unknown) weights, plus a small measurement error.

Low-vol regime
small noise ⇒ weights stay **stable**

High-vol regime
large noise ⇒ weights **adapt fast**

Predict–Update Cycle (every week)

- **Predict** — carry last week's weight estimate forward and widen uncertainty by the regime-set noise amount
- **Observe** — compare the predicted target return against what actually happened
- **Update** — correct the weight estimate proportionally to the forecast error; how much to correct is set by the *Kalman gain*, which balances trust in the model vs. trust in the new data

Model 2: Kalman Filter — Results

KF: Cumulative Performance vs Target



KF Performance Metrics

Metric	Rep.	Tgt.	Metric	Value
Ann. Return	**5.76%**	5.50%	Correlation	0.916
Ann. Volatility	**7.52%**	8.04%	Info. Ratio	+0.098
Sharpe Ratio	**0.77**	0.684	TEV	3.23%
Max Drawdown	**11.8%**	15.9%	Turnover	1.69

Advantages over MS-EN

- Updates **every week** (vs. every 4w for MS-EN)
- Inherently smoother weight paths via Kalman gain damping
- Lower turnover ⇒ lower implicit transaction costs
- No hyperparameter grid search — measurement noise calibrated by maximum likelihood on the training window

Key Takeaway

The KF achieves the **highest correlation** (0.916) and **lowest TEV** (3.23%) of all three models, making it the strongest pure replicator — at the cost of slightly higher volatility than MS-EN.

Model 3: Soft Bayesian Stacking — Best of Both Worlds

Sequential Likelihood-Weighted Combination

Instead of choosing one model the Stack **continuously tracks which model has been more accurate** and tilts toward it.

Each week, every model is scored on its accumulated log-likelihood, with past evidence down-weighted by a forgetting coefficient $\lambda = 0.97$. The model with the better recent track record receives a higher weight. The final forecast is then:

$$\hat{y}_t^{\star} = w_t^{\mathrm{MS}}\,\hat{y}_t^{\mathrm{MS}} + w_t^{\mathrm{KF}}\,\hat{y}_t^{\mathrm{KF}}$$

with weights summing to one and updated *before* seeing week t — no lookahead.

Design Choices

- **Bounded weights**: a hard floor of 15% and ceiling of 85% per model ensure diversification — neither model is ever silenced nor allowed to fully dominate

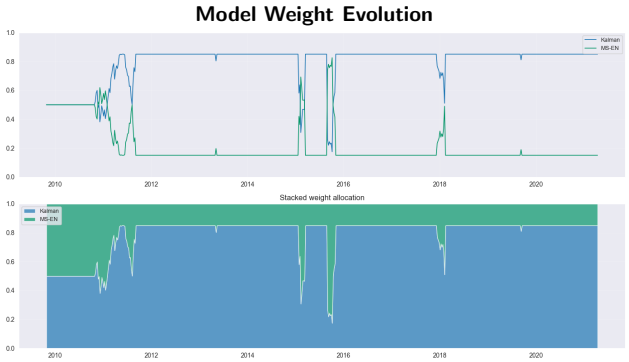
Stack: Cumulative Performance vs Target



Stack Performance Metrics

Metric	Replica	Target
Ann. Return	5.62%	5.50%
Ann. Volatility	7.23%	8.04%
Sharpe Ratio	0.78	0.684
Max Drawdown	11.7%	15.9%
Correlation	0.9136	
Information Ratio	+0.0546	
Tracking Error Vol	3.27%	
RMSE (weekly)	0.453%	
Ann. Turnover (meta)	0.5	

Model 3: Soft Bayesian Stacking — Stacking Weights



Average Stacking Weights

Model	Avg Weight
Kalman	0.7893
MS-EN	0.2107

Extended Evaluation: Multi-Dimensional Model Comparison



Reading the Table

Kalman tracks the target most tightly — best Corr (0.916), TEV (3.23%), MSE and Ann. Return (5.76%). It is the strongest *replicator*.

MS-EN manages risk most efficiently — best MaxDD (10.8%) and lowest volatility (6.85%).

Stack is a deliberate compromise: its value lies in *orthogonal failure coverage* — neither model alone covers the full spectrum of market regimes

Metric	MS-EN	KF	Stack	Metric	MS-EN	KF	Stack
Correlation	0.872	**0.916**	0.914	Max Drawdown	**10.8%**	11.8%	11.7%
TEV	3.94%	**3.23%**	3.27%	Ann. Return	5.36%	**5.76%**	5.62%
Inf. Ratio	−0.035	+0.098	+0.055	Ann. Volatility	**6.85%**	7.52%	7.23%
Sharpe	**0.78**	0.77	**0.78**	MSE	2.98e-05	**2.00e-05**	2.06e-05

Risk Analysis: VaR, Stress Test & Final Scoring

1 — Cornish-Fisher VaR (Regulatory)

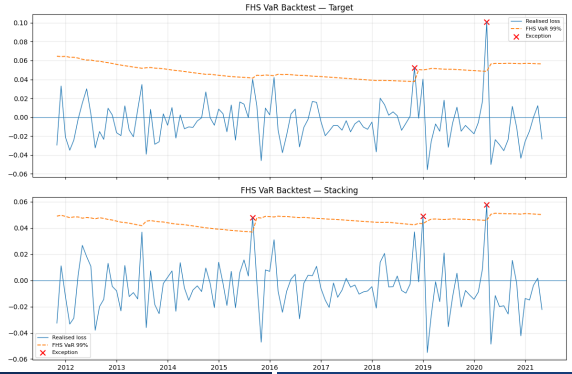
Adjusts the Normal quantile for skewness & excess kurtosis.

	Target	Stack
VaR 99%	7.26%	**5.04%**
ES 99%	9.14%	**6.08%**
Skew	−0.88	−0.46
Kurt	2.42	0.87

2 — FHS Backtest (Kupiec)

Expanding-window vol scaling; $LR_{UC} \sim \chi^2(1)$.

	Target	Stack		Target	Stack
Exceptions	2/115	3/115	Kupiec p	0.471	0.149
Exc. rate	1.74%	2.61%	Avg VaR 99%	4.90%	**4.55%**



3 — Stress VaR (Oct 2009 – Mar 2011)

Empirical quantile over first 18 OOS months
(EU sovereign crisis onset — GFC in IS window).

	Target	Stack
Stress VaR 99%	6.46%	**2.88%**
Stress ES 99%	7.21%	**3.07%**

Thank You

Questions & Discussion

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